

Real-network
RSRP/PCI traces



Temporal front end
LSTM / GRU / TCN



50-sample
RSRP windows



Grid-searched
HO classifier

Model selection: stratified train/test split, training-only grid search, F1-oriented selection, and held-out reporting

Reported metrics: accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices